

Chapter 2 Multivariate Statistics

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Plan

- 1 Geometry of the Sample
- 2 Random Samples
- 3 Generalized Variance
- 4 Sample Mean, Covariance, Correlation
- 5 Linear Combinations of Sample Values

X-Matrix

Data: p Variables/Characteristics Observed on n Experimental/Sampling Units:

$$\mathbf{X} = \begin{bmatrix} x_{11} & x_{12} & \cdots & x_{1p} \\ x_{21} & x_{22} & \cdots & x_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{np} \end{bmatrix} = \begin{bmatrix} \mathbf{x}_1' \\ \mathbf{x}_2' \\ \vdots \\ \mathbf{x}_n' \end{bmatrix} = \begin{bmatrix} \mathbf{y}_1 & \mathbf{y}_2 & \cdots & \mathbf{y}_p \end{bmatrix}$$

$$\mathbf{x}_j' = \begin{bmatrix} x_{j1} & x_{j2} & \cdots & x_{jp} \end{bmatrix} \quad j = 1, \dots, n$$

$$\mathbf{y}_i = \begin{bmatrix} x_{1i} \\ x_{2i} \\ \vdots \\ x_{ni} \end{bmatrix} \quad i = 1, \dots, p$$

Mean Vector

Mean Vector and Vector of a Single Mean:

$$\mathbf{x} = \begin{bmatrix} \bar{x}_1 \\ \bar{x}_2 \\ \vdots \\ \bar{x}_p \end{bmatrix} \quad \bar{x}_i = \frac{1}{n} \sum_{j=1}^n x_{ji} \quad i = 1, \dots, p \quad \bar{\mathbf{x}} = \frac{1}{n} \mathbf{X}' \mathbf{1}_n$$

$$\bar{x}_i \mathbf{1}_n = \begin{bmatrix} \bar{x}_i \\ \bar{x}_i \\ \vdots \\ \bar{x}_i \end{bmatrix} = \left[\mathbf{y}_i' \left(\frac{1}{\sqrt{n}} \mathbf{1}_n \right) \right] \left(\frac{1}{\sqrt{n}} \mathbf{1}_n \right) \quad i = 1, \dots, p$$

Projection of $\mathbf{y}_i$ on the vector $\mathbf{1}_n$ is $\bar{x}_i \mathbf{1}_n$ which has Length:

$$\sqrt{\bar{x}_i \mathbf{1}_n' \bar{x}_i \mathbf{1}_n} = |\bar{x}_i| \sqrt{n}.$$

Deviation Vectors

Vector of Deviations for a Single Variable, Sums of Squares and Cross - Products, and Correlation:

$$\mathbf{d}_i = \begin{bmatrix} x_{1i} - \bar{x}_i \\ x_{2i} - \bar{x}_i \\ \vdots \\ x_{ni} - \bar{x}_i \end{bmatrix} = \mathbf{y}_i - \bar{x}_i \mathbf{1}_n, i = 1, \dots, p \quad ; \quad \mathbf{d}_i' \mathbf{d}_i = \sum_{j=1}^n (x_{ji} - \bar{x}_i)^2$$

$$\mathbf{d}_i' \mathbf{d}_k = \sum_{j=1}^n (x_{ji} - \bar{x}_i) (x_{jk} - \bar{x}_k) = L_{\mathbf{d}_i} L_{\mathbf{d}_k} \cos(\theta_{ik})$$

$$\Rightarrow r_{ik} = \frac{\sum_{j=1}^n (x_{ji} - \bar{x}_i)(x_{jk} - \bar{x}_k)}{\sqrt{\sum_{j=1}^n (x_{ji} - \bar{x}_i)^2} \sqrt{\sum_{j=1}^n (x_{jk} - \bar{x}_k)^2}} = \cos(\theta_{ik})$$

$$\Rightarrow \theta_{ik} = \cos^{-1}(r_{ik})$$

Matrix of Deviations, Sum of Squares Cross - Products (SSCP)

$$\mathbf{E} = \begin{bmatrix} \mathbf{d}_1 & \mathbf{d}_2 & \cdots & \mathbf{d}_p \end{bmatrix} = \begin{bmatrix} x_{11} - \bar{x}_1 & x_{12} - \bar{x}_2 & \cdots & x_{1p} - \bar{x}_p \\ x_{21} - \bar{x}_1 & x_{22} - \bar{x}_2 & \cdots & x_{2p} - \bar{x}_p \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} - \bar{x}_1 & x_{n2} - \bar{x}_2 & \cdots & x_{np} - \bar{x}_p \end{bmatrix}$$

$$SSCP = \mathbf{E}'\mathbf{E} = \begin{bmatrix} \sum_{j=1}^n (x_{j1} - \bar{x}_1)^2 & \cdots & \sum_{j=1}^n (x_{j1} - \bar{x}_1) (x_{jp} - \bar{x}_p) \\ \sum_{j=1}^n (x_{j1} - \bar{x}_1) (x_{j2} - \bar{x}_2) & \cdots & \sum_{j=1}^n (x_{j2} - \bar{x}_2) (x_{jp} - \bar{x}_p) \\ \vdots & \ddots & \vdots \\ \sum_{j=1}^n (x_{j1} - \bar{x}_1) (x_{jp} - \bar{x}_p) & \cdots & \sum_{j=1}^n (x_{jp} - \bar{x}_p)^2 \end{bmatrix}$$

Matrix, and Sample Variance - Covariance Matrix ($\mathbf{S}_n$)

$$\mathbf{S}_n = \frac{1}{n} \mathbf{E}' \mathbf{E} = \frac{1}{n} \begin{bmatrix} \sum_{j=1}^n (x_{j1} - \bar{x}_1)^2 & \cdots & \sum_{j=1}^n (x_{j1} - \bar{x}_1) (x_{jp} - \bar{x}_p) \\ \sum_{j=1}^n (x_{j1} - \bar{x}_1) (x_{j2} - \bar{x}_2) & \cdots & \sum_{j=1}^n (x_{j2} - \bar{x}_2) (x_{jp} - \bar{x}_p) \\ \vdots & \ddots & \vdots \\ \sum_{j=1}^n (x_{j1} - \bar{x}_1) (x_{jp} - \bar{x}_p) & \cdots & \sum_{j=1}^n (x_{jp} - \bar{x}_p)^2 \end{bmatrix}$$

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Random Samples of Vectors of Units I

$$\mathbf{X} = \begin{bmatrix} X_{11} & X_{12} & \cdots & X_{1p} \\ X_{21} & X_{22} & \cdots & X_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ X_{n1} & X_{n2} & \cdots & X_{np} \end{bmatrix} = \begin{bmatrix} \mathbf{x}_1' \\ \mathbf{x}_2' \\ \vdots \\ \mathbf{x}_n' \end{bmatrix}$$

$$\mathbf{x}_j' = \begin{bmatrix} X_{j1} & X_{j2} & \cdots & X_{jp} \end{bmatrix} \quad j = 1, \dots, n$$

Row Vectors $\equiv$ independent observations from common joint probability distribution: $f(\mathbf{x}) = f(x_1, \dots, x_p)$

$\Rightarrow \mathbf{x}_1, \dots, \mathbf{x}_n$ can be treated as a Random Sample from $f(\mathbf{x})$

$\Rightarrow$ joint density is $f(\mathbf{x}_1) \cdots f(\mathbf{x}_n)$ where $f(\mathbf{x}_j) = f(x_{j1}, \dots, x_{jp})$.

Random Samples of Vectors of Units II

Treating Units (Rows) as a Random Sample from an Underlying Population of Units on the p Variables:

$$E \{X_{ji}\} = \mu_i$$

$$V \{X_{ji}\} = E \{(X_{ji} - \mu_i)^2\} = \sigma_{ii}$$

$$\text{Cov} \{X_{ji}, X_{jk}\} = E \{(X_{ji} - \mu_i)(X_{jk} - \mu_k)\} = \sigma_{ik}$$

Random Samples of Vectors of Units III

For Random Samples, Rows are Independent (Note: Not a Time - Series where rows are the same units being observed over Time):

$$\text{Cov} \{X_{ji}, X_{lk}\} = 0 \quad j \neq l, \forall i, k$$

$$E \{\mathbf{X}_j\} = \mu_{\mathbf{X}} = \begin{bmatrix} \mu_1 \\ \mu_2 \\ \vdots \\ \mu_p \end{bmatrix}$$

$$V \{\mathbf{X}_j\} = E \{(\mathbf{X}_j - \mu_{\mathbf{X}})(\mathbf{X}_j - \mu_{\mathbf{X}})'\} = \Sigma_{\mathbf{X}} = \begin{bmatrix} \sigma_{11} & \sigma_{12} & \cdots & \sigma_{1p} \\ \sigma_{21} & \sigma_{22} & \cdots & \sigma_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_{p1} & \sigma_{p2} & \cdots & \sigma_{pp} \end{bmatrix}$$

$$\text{Cov} \{\mathbf{X}_j, \mathbf{X}_l\} = E \{(\mathbf{X}_j - \mu_{\mathbf{X}})(\mathbf{X}_l - \mu_{\mathbf{X}})'\} = \mathbf{0}_{p \times p} \quad j \neq l$$

Expectation and Variance–Covariance Matrix of Sample Mean Vector

$$E \{ \mathbf{X}_j \} = \boldsymbol{\mu}_{\mathbf{X}} = \begin{bmatrix} \mu_1 \\ \mu_2 \\ \vdots \\ \mu_p \end{bmatrix}$$

$$V \{ \mathbf{X}_j \} = E \{ (\mathbf{X}_j - \boldsymbol{\mu}_{\mathbf{X}}) (\mathbf{X}_j - \boldsymbol{\mu}_{\mathbf{X}})' \} = \boldsymbol{\Sigma}_{\mathbf{X}} = \begin{bmatrix} \sigma_{11} & \sigma_{12} & \cdots & \sigma_{1p} \\ \sigma_{21} & \sigma_{22} & \cdots & \sigma_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_{p1} & \sigma_{p2} & \cdots & \sigma_{pp} \end{bmatrix}$$

$$\text{Cov} \{ \mathbf{X}_j, \mathbf{X}_l \} = E \{ (\mathbf{X}_j - \boldsymbol{\mu}_{\mathbf{X}}) (\mathbf{X}_l - \boldsymbol{\mu}_{\mathbf{X}})' \} = \mathbf{0}_{p \times p}$$

Expectation and Variance –Covariance Matrix of Sample Mean Vector

$$\text{Let } \bar{\mathbf{X}} = \frac{1}{n} (\mathbf{X}_1 + \dots + \mathbf{X}_n) = \frac{1}{n} \left(\begin{bmatrix} X_{11} \\ X_{12} \\ \vdots \\ X_{1p} \end{bmatrix} + \dots + \begin{bmatrix} X_{n1} \\ X_{n2} \\ \vdots \\ X_{np} \end{bmatrix} \right) = \begin{bmatrix} \bar{X}_1 \\ \bar{X}_2 \\ \vdots \\ \bar{X}_p \end{bmatrix}$$

$$E \{ \bar{\mathbf{X}} \} = E \left\{ \frac{1}{n} (\mathbf{X}_1 + \dots + \mathbf{X}_n) \right\} = \frac{1}{n} n E \{ \mathbf{X}_j \} = \mu_{\mathbf{X}}$$

$$\begin{aligned} V \{ \bar{\mathbf{X}} \} &= E \{ (\bar{\mathbf{X}} - \mu_{\mathbf{X}}) (\bar{\mathbf{X}} - \mu_{\mathbf{X}})' \} = E \left\{ \left[\frac{1}{n} \sum_{j=1}^n (\mathbf{X}_j - \mu_{\mathbf{X}}) \right] \left[\frac{1}{n} \sum_{l=1}^n (\mathbf{X}_l - \mu_{\mathbf{X}}) \right]' \right\} = \\ &= E \left\{ \frac{1}{n^2} \sum_{j=1}^n \sum_{l=1}^n (\mathbf{X}_j - \mu_{\mathbf{X}}) (\mathbf{X}_l - \mu_{\mathbf{X}})' \right\} = E \left\{ \frac{1}{n^2} \sum_{j=1}^n (\mathbf{X}_j - \mu_{\mathbf{X}}) (\mathbf{X}_j - \mu_{\mathbf{X}})' \right\} = \frac{1}{n} \Sigma_{\mathbf{X}} \end{aligned}$$

Expectation of Sample Variance-Covariance Matrix

$$\begin{aligned}
 \mathbf{S}_n &= \frac{1}{n} \mathbf{E}' \mathbf{E} = \frac{1}{n} \begin{bmatrix} \sum_{j=1}^n (x_{j1} - \bar{x}_1)^2 & \cdots & \sum_{j=1}^n (x_{j1} - \bar{x}_1) (x_{jp} - \bar{x}_p) \\ \sum_{j=1}^n (x_{j1} - \bar{x}_1) (x_{j2} - \bar{x}_2) & \cdots & \sum_{j=1}^n (x_{j2} - \bar{x}_2) (x_{jp} - \bar{x}_p) \\ \vdots & \ddots & \vdots \\ \sum_{j=1}^n (x_{j1} - \bar{x}_1) (x_{jp} - \bar{x}_p) & \cdots & \sum_{j=1}^n (x_{jp} - \bar{x}_p)^2 \end{bmatrix} = \\
 &= \frac{1}{n} \sum_{j=1}^n [(\mathbf{x}_j - \bar{\mathbf{x}}) (\mathbf{x}_j - \bar{\mathbf{x}})'] = \frac{1}{n} \sum_{j=1}^n [\mathbf{x}_j \mathbf{x}_j' - \mathbf{x}_j \bar{\mathbf{x}}' - \bar{\mathbf{x}} \mathbf{x}_j' + \bar{\mathbf{x}} \bar{\mathbf{x}}'] \\
 &= \frac{1}{n} \sum_{j=1}^n \mathbf{x}_j \mathbf{x}_j' - \bar{\mathbf{x}} \bar{\mathbf{x}}' - \bar{\mathbf{x}} \bar{\mathbf{x}}' + \bar{\mathbf{x}} \bar{\mathbf{x}}' = \frac{1}{n} \sum_{j=1}^n \mathbf{x}_j \mathbf{x}_j' - \bar{\mathbf{x}} \bar{\mathbf{x}}'
 \end{aligned}$$

Expectation of Sample Variance-Covariance Matrix

Aside: Let $\mathbf{V}$ be a random vector with mean vector and Variance - Covariance matrix: $\mu_{\mathbf{V}}$ and $\Sigma_{\mathbf{V}}$:

$$\begin{aligned}\Sigma_{\mathbf{V}} &= E \{ (\mathbf{V} - \mu_{\mathbf{V}}) (\mathbf{V} - \mu_{\mathbf{V}})' \} = E \{ \mathbf{V}\mathbf{V}' - \mathbf{V}\mu_{\mathbf{V}}' - \mu_{\mathbf{V}}\mathbf{V}' + \mu_{\mathbf{V}}\mu_{\mathbf{V}}' \} = \\ &= E \{ \mathbf{V}\mathbf{V}' \} - \mu_{\mathbf{V}}\mu_{\mathbf{V}}' - \mu_{\mathbf{V}}\mu_{\mathbf{V}}' + \mu_{\mathbf{V}}\mu_{\mathbf{V}}' = E \{ \mathbf{V}\mathbf{V}' \} - \mu_{\mathbf{V}}\mu_{\mathbf{V}}' \\ \Rightarrow E \{ \mathbf{V}\mathbf{V}' \} &= \Sigma_{\mathbf{V}} + \mu_{\mathbf{V}}\mu_{\mathbf{V}}' \\ \Rightarrow E \{ \mathbf{S}_n \} &= E \left\{ \frac{1}{n} \sum_{j=1}^n \mathbf{x}_j \mathbf{x}_j' - \bar{\mathbf{x}} \bar{\mathbf{x}}' \right\} = \\ &= \left[\frac{1}{n} n (\Sigma_{\mathbf{X}} + \mu_{\mathbf{X}} \mu_{\mathbf{X}}') \right] - \left[\frac{1}{n} \Sigma_{\mathbf{X}} + \mu_{\mathbf{X}} \mu_{\mathbf{X}}' \right] = \frac{n-1}{n} \Sigma_{\mathbf{X}}\end{aligned}$$

Defining:

$$\mathbf{S} = \frac{n}{n-1} \mathbf{S}_n = \frac{1}{n-1} \sum_{j=1}^n [(\mathbf{x}_j - \bar{\mathbf{x}}) (\mathbf{x}_j - \bar{\mathbf{x}})'] \Rightarrow E \{ \mathbf{S} \} = \Sigma_{\mathbf{X}}.$$

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Generalized Variance I

$$\mathbf{S} = \begin{bmatrix} s_{11} & s_{12} & \cdots & s_{1p} \\ s_{12} & s_{22} & \cdots & s_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ s_{1p} & s_{2p} & \cdots & s_{pp} \end{bmatrix} \quad s_{ik} = \frac{1}{n-1} \sum_{j=1}^n (x_{ji} - \bar{x}_i)(x_{jk} - \bar{x}_k)$$

Generalized Sample Variance = $\det(\mathbf{S}) = |\mathbf{S}|$.

$$p = 2: \quad \mathbf{S} = \begin{bmatrix} s_{11} & s_{12} \\ s_{12} & s_{22} \end{bmatrix} = \begin{bmatrix} s_{11} & r_{12}\sqrt{s_{11}}\sqrt{s_{22}} \\ r_{12}\sqrt{s_{11}}\sqrt{s_{22}} & s_{22} \end{bmatrix}$$

$$\Rightarrow |\mathbf{S}| = s_{11}s_{22} - r_{12}^2 s_{11}s_{22} = s_{11}s_{22} (1 - r_{12}^2) =$$

$$= s_{11}s_{22} (1 - \cos^2 \theta_{12}) = s_{11}s_{22} \sin^2 \theta_{12}.$$

Generalized Variance II

Area of Trapezoid created from vectors $\mathbf{d}_1, \mathbf{d}_2$ (plot on next slide):

$$\text{Area} = L_{\mathbf{d}_1} L_{\mathbf{d}_2} |\sin \theta_{12}|$$

with

$$\theta_{12} = \cos^{-1} r_{12}$$

$$L_{\mathbf{d}_1} = \sqrt{\sum_{j=1}^n (x_{j1} - \bar{x}_1)^2} = \sqrt{(n-1) s_{11}}$$

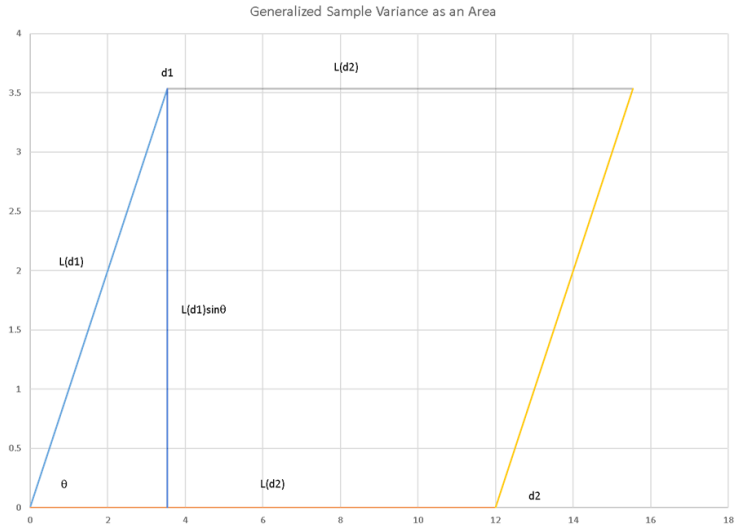
$$L_{\mathbf{d}_2} = \sqrt{\sum_{j=1}^n (x_{j2} - \bar{x}_2)^2} = \sqrt{(n-1) s_{22}}.$$

$$\Rightarrow \text{Area} = (n-1) \sqrt{s_{11} s_{22}} |\sin \theta_{12}|$$

$$\Rightarrow |\mathbf{S}| = \frac{\text{Area}^2}{(n-1)^2}$$

For general p: $|\mathbf{S}| = \frac{\text{Volume}^2}{(n-1)^p}$

Volume of a hyperellipsoid centered at $\bar{\mathbf{x}}$:



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Matrix Form of Sample Mean

$$\bar{\mathbf{x}} = \begin{bmatrix} \bar{x}_1 \\ \bar{x}_2 \\ \vdots \\ \bar{x}_p \end{bmatrix} = \frac{1}{n} \mathbf{X}' \mathbf{1}_n$$

$n \times p$ matrix of Column means:

$$\begin{bmatrix} \bar{x}_1 \mathbf{1}_n & \bar{x}_2 \mathbf{1}_n & \cdots & \bar{x}_p \mathbf{1}_n \end{bmatrix} = \mathbf{1}_n \bar{\mathbf{x}}' = \frac{1}{n} \mathbf{1}_n \mathbf{1}_n' \mathbf{X} = \frac{1}{n} \mathbf{J}_n \mathbf{X}$$

Note: $\frac{1}{n} \mathbf{J}_n \frac{1}{n} \mathbf{J}_n = \frac{1}{n^2} \mathbf{J}_n \mathbf{J}_n = \frac{1}{n^2} n \mathbf{J}_n = \frac{1}{n} \mathbf{J}_n$.

$n \times p$ matrix of Deviations from Column means:

$$\mathbf{E} = \mathbf{X} - \frac{1}{n} \mathbf{J}_n \mathbf{X} = \left(\mathbf{I}_n - \frac{1}{n} \mathbf{J}_n \right) \mathbf{X}$$

Matrix Form of Sample Covariance, Correlation

Note: $\left(\mathbf{I}_n - \frac{1}{n}\mathbf{J}_n\right)\left(\mathbf{I}_n - \frac{1}{n}\mathbf{J}_n\right) = \mathbf{I}_n - \frac{1}{n}\mathbf{J}_n - \frac{1}{n}\mathbf{J}_n + \frac{1}{n^2}\mathbf{J}_n\mathbf{J}_n = \mathbf{I}_n - \frac{1}{n}\mathbf{J}_n$.

$p \times p$ matrix of Sums of Squares and Cross - Products and Estimated Variance - Covariance Matrix:

$$\mathbf{E}'\mathbf{E} = \mathbf{X}'\left(\mathbf{I}_n - \frac{1}{n}\mathbf{J}_n\right)'\left(\mathbf{I}_n - \frac{1}{n}\mathbf{J}_n\right)\mathbf{X} = \mathbf{X}'\left(\mathbf{I}_n - \frac{1}{n}\mathbf{J}_n\right)\mathbf{X}$$

$$\Rightarrow \mathbf{S} = \frac{1}{n-1}\mathbf{E}'\mathbf{E} = \frac{1}{n-1}\mathbf{X}'\left(\mathbf{I}_n - \frac{1}{n}\mathbf{J}_n\right)\mathbf{X}.$$

$p \times p$ Standard Deviation Matrix and its Inverse:

$$\mathbf{D}^{1/2} = \begin{bmatrix} \sqrt{s_{11}} & 0 & \cdots & 0 \\ 0 & \sqrt{s_{22}} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \sqrt{s_{pp}} \end{bmatrix}$$

$$\mathbf{D}^{-1/2} = \begin{bmatrix} 1/\sqrt{s_{11}} & 0 & \cdots & 0 \\ 0 & 1/\sqrt{s_{22}} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1/\sqrt{s_{pp}} \end{bmatrix}$$

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Linear Combinations of Sample Values

$$\mathbf{X} = \begin{bmatrix} X_1 \\ X_2 \\ \vdots \\ X_p \end{bmatrix} ; \quad \mathbf{c}'\mathbf{X} = c_1X_1 + \dots + c_pX_p$$

observed value for Unit j : $\mathbf{c}'\mathbf{x}_j = c_1x_{j1} + \dots + c_px_{jp} \quad j = 1, \dots, n$

Sample Mean: $\frac{\mathbf{c}'\mathbf{x}_1 + \dots + \mathbf{c}'\mathbf{x}_n}{n} = \frac{1}{n}\mathbf{c}'(\mathbf{x}_1 + \dots + \mathbf{x}_n) = \mathbf{c}'\bar{\mathbf{x}}$

Sample Variance: $\frac{1}{n-1} \sum_{j=1}^n (\mathbf{c}'\mathbf{x}_j - \mathbf{c}'\bar{\mathbf{x}})^2 = \frac{1}{n-1} \sum_{j=1}^n [\mathbf{c}'(\mathbf{x}_j - \bar{\mathbf{x}})]^2 =$

$$= \frac{1}{n-1} \sum_{j=1}^n [\mathbf{c}'(\mathbf{x}_j - \bar{\mathbf{x}})(\mathbf{x}_j - \bar{\mathbf{x}})'\mathbf{c}] = \mathbf{c}' \frac{\sum_{j=1}^n (\mathbf{x}_j - \bar{\mathbf{x}})(\mathbf{x}_j - \bar{\mathbf{x}})'}{n-1} \mathbf{c} = \mathbf{c}'\mathbf{S}\mathbf{c}$$

Linear Combinations of Sample Values

Second Linear Combination: $\mathbf{b}'\mathbf{X}$

Sample Mean: $\mathbf{b}'\bar{\mathbf{x}}$ and Sample Variance: $\mathbf{b}'\mathbf{S}\mathbf{b}$

Sample Variance of $\mathbf{b}'\mathbf{X}$ and $\mathbf{c}'\mathbf{X}$: $\frac{1}{n-1} \sum_{j=1}^n [\mathbf{b}'(\mathbf{x}_j - \bar{\mathbf{x}}) \mathbf{c}'(\mathbf{x}_j - \bar{\mathbf{x}})] =$

$$\frac{1}{n-1} \sum_{j=1}^n [\mathbf{b}'(\mathbf{x}_j - \bar{\mathbf{x}})(\mathbf{x}_j - \bar{\mathbf{x}})' \mathbf{c}] = \mathbf{b}' \frac{\sum_{j=1}^n (\mathbf{x}_j - \bar{\mathbf{x}})(\mathbf{x}_j - \bar{\mathbf{x}})'}{n-1} \mathbf{c} = \mathbf{b}'\mathbf{S}\mathbf{c}$$

Generalized to q Linear Combinations:

$$a_{i1}X_1 + \dots + a_{ip}X_p \quad i = 1, \dots, q \quad \mathbf{A}\mathbf{X} \text{ where } \mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1p} \\ a_{21} & a_{22} & \cdots & a_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ a_{q1} & a_{q2} & \cdots & a_{qp} \end{bmatrix}$$

Sample Mean: $\mathbf{A}\bar{\mathbf{x}}$ Sample Variance: $\mathbf{A}\mathbf{S}\mathbf{A}'$